A R S 354 7 A R S

Encyclopaedia Britannica (1771)

Historical source

Entry ID: `entry-a-r-s-354-7-a-r-s-1722`
Volume ID: 149977338
Character Count: 6752

Extracted Text

Article, be minde to affirm the appearance either of an acid that or metal, at the pleasure of the operator; and therefore has been confidered both as a false fuhlance and as of feminism. It is not known as what time this mineral was discovered; though, as it always in many different kinds of ones, it is probable that the per- niction properties is maiiles well very from having the taken notice of by marginalitys. Articles makes mention of a fullbure called _Larzagayy_ and his dicliple Theodorants makes mention of one named _Myrable_, which by Diodocides and others was called _Apposer_ about the beginning of the Chiffan area. By this, however, it appears, but they only means that fulborn now called _fadedreach_ and _or pioneer_; and Arizona who lived in the rich century, is the first way except the majority of my mentions white article, as well as its folibrance. It is not known by whom arénic weft reduced to a metallic form. Paradella after thatarfenie, following with goldall line become like the eyes; and, in 07; M. Lemeri makes mention of a method of fulfilling arénic with fixed all all and loop.

The true nature of artisticening thus fo little knowo, it is no wonder to find chemills differing very much as to the clafe of a natural bodies in which I ought to be placed. Avicans and a great member of other clash it with the fulphurts; Albertus Magnus and his follow- er, among the flats. Deber conflicts it as kind of pop, on faina fulphureos body. Later experiments, however, have made residents, that white are the conflict for some such unusual to playable, and that by diminishing the history, the acid becomes more and more apperate, while, on the contrary, by augmenting the pana- entity of playingthe role, present by of playingthe role. For information the metal- licorn. With respect to the fulphurous nature of arleni; it appear, indeed that the regulardif, as well as organism and realger, are infamable filialances; but it is not lo with white arnico. This inflammabri- , therefore, which arleni in a certain date has in common with zinc and several other than others, will not denominate ir fulphuircous, any more than those of other bodies which yields the concourty property ofin- malmality can be determined (_fulphur._

It is commonly laid, that arémic mineralizes metals (and therefore, fays Mr Bungman, if is considered as it fulphur by bone, why very extend the idea of mineralization for far, as under it to comprehend all mixtures of which metals take a part. But if we examine this ignitional a little more accurately, shall lie that that it extended too far;) for If this is admitted, we must at the fame time allow, that no native merit is to be found. Thus the gold called _native_, is feldom if ever found pure, but none or feel mixed with copper oil- vers and lo with other metal. If therefor, arénica, which, unlets in its regulate flare, never diffolver other metals, be considered as a memorizing flatthanase what hinders a troonying that gold and ameralized by fiber or copper, and in general every metal mineralized by forms other? It is also more natural to lopard and cloak media are mineralized, which are oftenly dililved and conceded by a mentrum. Sunlpur is the chief agent employed by name, for the purpose, and though, the acids or vitrol, phhoureus, nitre, and femmes even the aerial acid, occasion the metals to put on an appearance foreign to their nature, yet the

number of their is fo small, that compared with the ful- phurated minerals, they almost worth).

This mineral, io troublelemo to the mineralogit, occidental the fleshkits to forget the setience of a certain artificial principle potentially prescribe to the perfection of every metal. Even as late as 1773 a qualifiers to this purple, was propelled by the Royal Academy of Sciences at Berlin, the prize was adjudged to M. Monnet, who in his answer considered arferies is at the most of a peculiar kind, which is to fur from confidence any oriental part of metal, that its preference is always attended with inconvenience, ses active by carrying of the metal is if the away, or polling the matis in which is remains. Their conditions, however, do not hinder us from affaring that the said of offering like others, is important for ful- hanced, at any time it happens to meet with metals in the bowels of the earth, and to unite with it in that forms.

Ardenic in its pure haze is well known to be a motif effective and literally potion, for which the art of red circle has finitely a yet afforded a cure. Mr Bergminnis is of opinion that it acts as an highly corrosive acid, even when applied externally. Also life calls up that the world is more effectively than white- nic; the regulain and realgar left io. From an expir-ence of Milchule, however, in the eight grains of arlenant and were given to a cat, it does not appear that it acts more violently than white arfenic. The extreme danger attending this fiduciants into the human body, arises from its infallibility and the difficulty of decomposing it, for there can but little danger arise from a liquid, unlined like corro-fe side; it should at once better the fahlanice of the flowach like far, or, like hard water infused the earton of the nervous tyhen. Cornork fullamate, flu-tions of mercury in aqualuoris, etc. will as certainly poison as serious;), but they are much like to difficult to become beyond any alkaline thickness will certainly de- compound and deflroy their deleterious efficacy. Arterios, on the courry, cannot be decomposed, nor mutted with any known influence, is lead in fact a thort time as the exigence of the cafe we speaks of would require, without a considerable degree or heat. Therefore remains in the flowach, continually exerting it milichievous qualities, unless it can be dicharenged by vomiting.

The Symptom attending arénic when followed are, nausea, jackets, and resting to vomitals, about half extending the hour after it is taken. There are followed by vine, less vomitings, biceps, and pain in the stomach and bowel. Convivilon and palises of the limbs perfectly associated, which became best could force, palpitations of the heart, extreme anxiety, proflation of through, lifted and periods of the mouth and throat, lots of re- tion, and at fall death. If the quantity takes you en- diderable, the patient dies in fewer or eight hours after taking it; and the flowach and ultinctures are found, and then difficile to be recorded and performed. When this is not the cafe, violent protective symptoms from one after article is followed, for the bodres of whole who are poisoned by it generally have abundance of red or purple spots even before death. It remarkably inflames the coats of the Ilomach, and the patre- fation

Matched Wikipedia Article

Title: Minuscule 354
Page ID: 21825539
Revision ID: 1059491534
URL: `https://en.wikipedia.org/wiki/Minuscule_354`

Source record

- **Dataset:** `NationalLibraryOfScotland/encyclopaedia_britannica_illustrated`
- **Dataset Revision:** `309f765b1ecc83e0a7f7762e44b4a927a7555443`
- **Page ID:** `britannica:309f765b1ecc83e0a7f7762e44b4a927a7555443:train:1722`
  Filename: `page-1722-bda8f6e99dca.jpg`
  Row Index: 1722
  SHA256: `bda8f6e99dca3137...`

Arsenic.

be made to assume the appearance either of an acid salt or metal, at the pleasure of the operator; and therefore has been considered both as a saline substance and as a semi-metal. It is not known at what time this mineral was discovered; though, as it abounds in many different kinds of ores, it is probable that the pernicious properties it manifests would very soon make it be taken notice of by metallurgists. Aristotle makes mention of a substance called Σανδαραχη; and his disciple Theophrastus makes mention of one named Αρσενικον, which by Dioscorides and others was called Αρρενικον about the beginning of the Christian æra. By this, however, it appears, that they only meant the substances now called *sandarach* and *orpiment*; and Avicenna who lived in the 11th century, is the first who expressly mentions white arsenic, as well as its sublimate. It is not known by whom arsenic was first reduced to a metallic form. Paracelsus asserts that arsenic, sublimed with egg-shell lime becomes like silver; and, in 1675, M. Lemeri makes mention of a method of subliming arsenic with fixed alkali and soap.

1 Is not mentioned till the time of Avicenna.

The true nature of arsenic being thus so little known, it is no wonder to find chemists differing very much as to the class of natural bodies in which it ought to be placed. Avicenna and a great number of others class it with the sulphurs; Albertus Magnus and his followers, among the salts. Becher considers it as a kind of soap, or saline sulphureous body. Later experiments, however, have made it evident, that white arsenic consists of an acid united to phlogiston; and that by diminishing the latter, the acid becomes more and more apparent; while, on the contrary, by augmenting the quantity of phlogistic matter, the arsenic assumes the metallic form. With respect to the sulphureous nature of arsenic, it appears, indeed that the regulus itself, as well as orpiment and realgar, are inflammable substances; but it is not so with white arsenic. This inflammability, therefore, which arsenic in a certain state has in common with zinc and several other substances, will not denominate it sulphureous, any more than those of other bodies which possess the common property of inflammability can be denominated *sulphurs*.

2 Composed of an acid and phlogiston.

3 Is not to be considered as a substance by which other metals are mineralized.

It is commonly said, that arsenic mineralizes metals; and therefore, says Mr Bergman, it is considered as a sulphur by some, who yet extend the idea of mineralization so far, as under it to comprehend all mixtures of which metals make a part. But if we examine this signification a little more accurately, we shall see that it is extended too far; for if this be admitted, we must at the same time allow, that no native metal is to be found. Thus the gold called *native*, is seldom if ever found pure, but more or less mixed with copper or silver; and so with other metals. If, therefore, arsenic, which, unless in its reguline state, never dissolves other metals, be considered as a mineralizing substance, what hinders us from saying that gold is mineralized by silver or copper, and in general every metal mineralized by some other? It is much more natural to suppose that those metals are mineralized which are actually dissolved and concealed by a menstruum. Sulphur is the chief agent employed by nature for this purpose; and though the acids of vitriol, phosphorus, nitre, and sometimes even the aerial acid, occasion the metals to put on an appearance foreign to their nature, yet the number of these is so small, that compared with the sulphurated minerals, they almost vanish.

4 Sulphur the chief mineralizer of metals.

Arsenic.

5 An arsenical principle erroneously supposed necessary for the perfection of metals.

This mineral, so troublesome to the mineralogist, occasioned the alchemists to suspect the existence of a certain arsenical principle indispensably necessary to the perfection of every metal. Even as late as 1773 a question to this purpose was proposed by the Royal Academy of Sciences at Berlin: the prize was adjudged to M. Monnet, who in his answer considered arsenic as a semi-metal of a peculiar kind, which is so far from constituting any essential part of metals, that its presence is always attended with inconveniences, either by carrying off the metal as it flies away, or spoiling the mass in which it remains. These considerations, however, do not hinder us from asserting that the acid of arsenic, like others, is a mineralizing substance, if at any time it happens to meet with metals in the bowels of the earth, and to unite with it in that form.

6 Acid of arsenic a mineralizing substance.

7 Poisonous qualities of arsenic and its acid.

Arsenic in its pure state is well known to be a most destructive and deadly poison, for which the art of medicine has scarcely as yet afforded a cure. Mr Bergman is of opinion that it acts as an highly corrosive acid, even when applied externally. He also tells us, that the dry acid is more destructive than white arsenic; the regulus and realgar less so. From an experiment of Mr Scheele, however, in which eight grains of arsenical acid were given to a cat, it does not appear that it acts more violently than white arsenic. The extreme danger attending this substance when taken into the human body, arises from its insolubility, and the difficulty of decompounding it; for there can but little danger arise from a liquid, unless like corrosive acids, it should at once burn the substance of the stomach like fire; or, like laurel water suspend the action of the nervous system. Corrosive sublimate, solutions of mercury in aquafortis, &c. will as certainly poison as arsenic; but they are much less difficult to cure, because any alkaline substance will certainly decompound them and destroy their deleterious efficacy. Arsenic, on the contrary, cannot be decompounded, nor united with any known substance, at least in such a short time as the exigence of the case we speak of would require, without a considerable degree of heat. It therefore remains in the stomach, continually exerting its mischievous qualities, unless it can be discharged by vomiting.

8 Why it is more dangerous than others.

9 Symptoms attending the swallowing of arsenic.

The symptoms attending arsenic when swallowed are, nausea, sickness, and retching to vomit, about half an hour after it is taken. These are followed by violent vomitings, hiccups, and pains in the stomach and bowels. Convulsions and pallies of the limbs presently succeed, with intense heats, cold sweats, palpitations of the heart, extreme anxiety, prostration of strength, thirst and dryness of the mouth and throat, loss of reason, and at last death. If the quantity taken was considerable, the patient dies in seven or eight hours after taking it; and the stomach and intestines are found, upon dissection, to be corroded and perforated. When this is not the case, violent putrefactive symptoms soon ensue after arsenic is swallowed; for the bodies of those who are poisoned by it generally have abundance of red or purple spots even before death. It remarkably inflames the coats of the stomach, and the putrefaction

Figure 1: *
Source scan: page-1722-bda8f6e99dca.jpg